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[18CE733]

SEVENTH SEMESTER B.Tech EXAMINATION, DEC 2023

PAVEMENT DESIGN

Time: 3 Hours **Maximum Marks: 70**

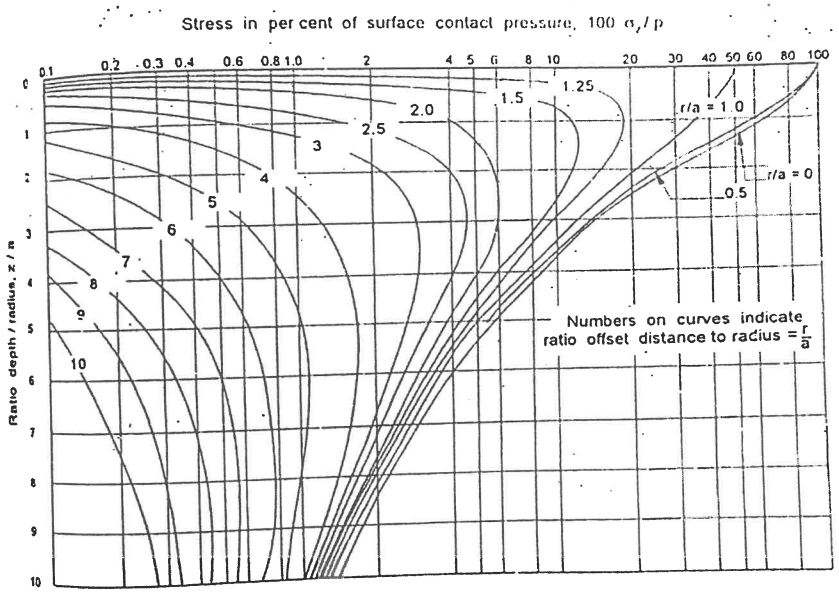
Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Describe in detail about rigid pavement with neat sketches. **09**

b. What are the advantages of providing flexible pavement. **05**

2. a. Describe the process for the plate bearing test. What are the corrections that need to be made to the modulus of subgrade reaction? **09**

b. A circular load of radius 15 cm with uniform contact pressure of 7.0 kg/cm^2 is applied on the surface of a homogeneous elastic mass. Determine the vertical stress under the centre of the load at a depth of 45 cm from the surface. **05**



3. Soil sub-grade sample obtained from the site was undergone CBR test in laboratory which gives 2.5 mm penetration is 60 kg and 5.00 mm penetration is 82 kg. **14**

Materials used over soil subgrade are:

a) Compacted sandy soil of 6% CBR
b) Poorly graded gravel of 12% CBR
c) Well graded gravel of 60% CBR
d) Thickness of bituminous concrete surfacing is 4 cm.

Design the pavement using CBR method. Assume wheel load as 4085 kg and tyre

pressure as 7 kg/cm^2.

OR

4. a. Calculate the pavement thickness (in cm) of base course by triaxial test method using the following data: **05**

Wheel load = 4100 kg, Radius of contact area = 15 cm, Traffic coefficient X = 1.5, Rainfall coefficient Y = 0.9, Design deflection Δ = 0.25 cm, E value of subgrade soil = 100 kg/cm^2, E value of base course material = 400 kg/cm^2.

b. Explain the design strategy for flexible pavement using the Mcleod method. **09**

5. A 50 kN wheel load is applied on a slab of 250 mm thickness. The radius of the wheel contact area is 160 mm and the radius of the relative stiffness value is 1.10 m. Determine the maximum stress when load at the corner, edge and interior locations of a slab Westergaard equation. **14**

OR

6. a. Describe the various types of joints found in rigid pavement. **10**

b. The design thickness of a CC pavement is 25 cm and the spacing between longitudinal joints is 3.5 m. If the allowable tensile stress in concrete during the initial curing period is 0.8 kg/cm^2, determine the spacing between the contraction joints in order to prevent random cracking across the CC pavement. The unit weight of CC = 2400 kg/m^3 and the friction coefficient between the membrane supporting layer and bottom of CC slab = 1.2. **04**

7. a. What are the common reasons for pavement failure? **05**

b. Explain the concept of Present Serviceability Index (PSI). **09**

OR

8. Benkelman beam deflection studies were carried out on 10 selected points on a stretch of a flexible pavement during summer season using a dual wheel load of 4085 kg, 5.6 kg/cm^2 pressure. The deflection values obtained in mm after making the necessary lag corrections are given below. **14**

1.22, 1.4, 1.33, 1.5, 1.6, 1.55, 1.25, 1.36, 1.62, 1.48.

If the present traffic consists of 750 commercial vehicles per day, determine the thickness of bituminous overlay required, if the pavement temperature during the test was 30° C and the correction factor for subsequent increase in subgrade moisture content is 1.4. Assume annual rate of growth of traffic as 8%. Adopt IRC guidance.